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Empowering Accessibility: Human-Centered Approach to a BCI Home Control for Impaired People

Júlia Ramos[1,2], Samuel Silva[1], Bernardo Marques[1], Miguel Pais-Vieira[2], Andrew Stevenson[3] and Susana Brás[1]

***Abstract*— Brain-Computer Interfaces (BCIs) have shown significant potential for individuals with motor impairments, either by improving physiotherapy treatments or by enabling to perform simple tasks, autonomously. However, much of this progress remains confined to controlled laboratory environments. This study aims to develop a BCI-controlled interface, for real-life scenario, tailored to allow individuals with Locked-In Syndrome (LIS) to interact with their home environment. To ensure system usability, a Human-Centered Design (HCD) approach was adopted prioritizing end-user needs. The interface control system was tested using a BITalino for Electroencephalogram (EEG) acquisition. Preliminary results demonstrated that professionals recognize the system's potential, highlighting the importance of real-time feedback, and design simplicity features to minimize user fatigue and improve control accuracy.**

***Clinical Relevance*—This interdisciplinary methodology bridges the gap between assistive technologies and the user needs, promoting autonomy and communication with a BCI-controlled interface for real home interaction.**

***Keywords*—Human-Centered Design; Brain-Computer Interfaces; Assistive Technology; Locked-in Syndrome**

I. INTRODUCTION

Brain-Computer Interfaces (BCIs) or Brain-Machine Interfaces (BMIs) are a promising avenue to restore the pathway between brain electrical commands and an external device control, allowing human capacity augmentation or impaired people to regain autonomy. The main goal of these interfaces is to decode brain activity and understand neural patterns to command hardware and software applications. BMIs are mainly composed by five consecutive stages: (1) signal acquisition of brain activity; (2) pre-processing methods to eliminate undesired interference (noise) (3) feature extraction to identify discriminative valuable data without losing crucial information (4) classification algorithms that try to find neural patterns in data, and (5) control commands for appliances, such as a wheelchair or a computer [1], as illustrated in Fig. 1.

Although there is still much to discover about the brain and how to interpret it, several paradigms can be used to control a BCI depending on the application. The most common neural control signals include Sensorimotor Rhythms (SMRs) located in the sensorimotor cortex, used in Motor Imagery (MI) tasks where participants imagine performing

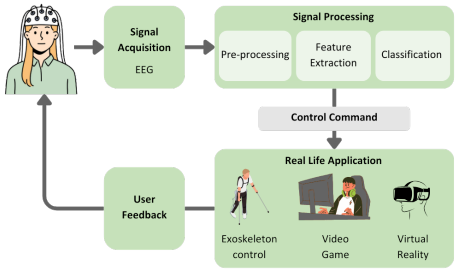


Fig. 1. Fundamental components of a standard BCI system

a motor movement, such as grasping an object. Steady-State Visual Evoked Potentials (SSVEPs) control paradigm can be detected in brain activity when a visual stimulus blinks at a specific frequency, and the Event-Related Potentials (ERPs) which include the P300 that correspond to EEG positive peaks that appear 300 ms after a certain stimulus is given to the subject [1].

Empowered by the great impact that BCIs have in improving a patient's life, the number of publications has grown in the last years [2], as Fig. 2 shows. Despite this evolution, the main part of this research is conducted in laboratory environments with healthy subjects, where the main focus remains on the system performance. Therefore, research involving end-users in home environments without expert supervision is still scarce, particularly due to the inherent problematics that arise when working with impaired patients.

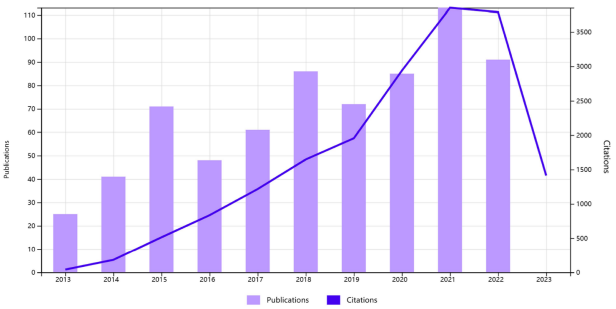


Fig. 2. Publication and citation trends over the last years for BCIs (figure retrieved from [2]).

The study proposed by Schreuder et al. [4] explores a user-centered design approach to develop a BCI system, testing multiple BCI paradigms to find the most suitable for the participant enrolled in the study, a woman who suffered

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from a stroke. This study concluded that visual paradigms were more suitable for the subject with lower cognitive workload while the auditory paradigm was not successful. Work like the one developed by Wong et al. [13] focuses on improving algorithms and decoding performances. Although these studies have led to significant advancements in the field, they still lack a fully autonomous control dimension. This includes essential functions such as turning on the screen, managing the interface, locking the system, and reactivating it when needed.

Traditional BCI research often focuses on neuroplasticity for rehabilitation purposes and the performance accuracy of Machine Learning (ML) models for EEG decoding, limiting the applicability of the technology to different real-world scenarios. Offering progress to what is more commonly found in literature, this paper studies the integration of a BCI to control a real home environment, involving health professionals in the design process.

This study highlights the importance of a Human-Centered Design (HCD) approach in developing assistive applications such as a BCI-controlled interface, ensuring that the technology design and interaction adapts to user needs [3]. By tailoring the system to disabled individuals, key features such as environmental complexity, user comfort, adaptability and scalability need to be considered. This study highlights the positive impact of an iterative design process that incorporates paper prototyping to improve usability before digital implementation, promoting the communication between end-users and developers. This hands-free interface enables and empowers individuals with severe physical limitations to independently control their environment.

The paper is organized as follows, Section II describes the methods and materials, Section III presents the implementation challenges while proposing the future guidelines for this project, and final remarks are presented in Section IV.

II. METHODS

A HCD approach adapted to BCI systems, Fig. 3, was embraced to align the system functionality with the user's needs and capabilities. A crucial aspect to successfully implement this approach is to involve the user in the design and evaluation process, therefore the first step is to comprehend their lifestyle and how to improve their quality, this can be accessed by using interviews or questionnaires [3][6].

International Organization for Standardization (ISO) 9241-210:2019 [7] provides guidelines for a HCD in interactive systems, indicating 4 crucial phases, identification of user needs, design development addressing those needs, prototype development and, evaluation of the developed system based on user feedback.

A. Implementation Challenges

The present study proposes a Human-Centered approach to design a Graphical User Interface (GUI) controlled by a SSVEP-BCI. However, the implementation of assistive systems as such comes with challenges that must be considered.

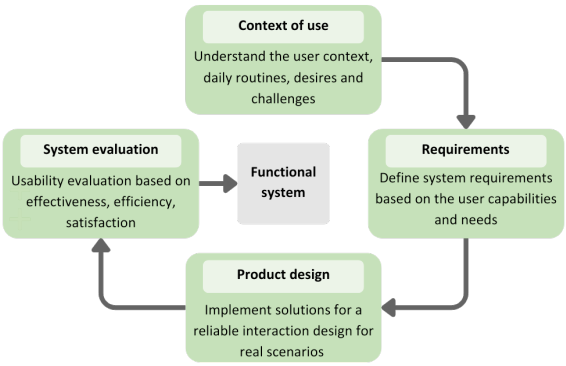


Fig. 3. Human-centered design main stages adapted to BCI applications.

- Real-time EEG signal decoding is a complex process, partially due to interferences during signal acquisition. Due to noise and signal complexity, identifying patterns in neural activity is a demand task that imposes accurate signal processing techniques, reliable recording EEG systems and electrode stability over time [2].
- Usability evaluation in terms of system effectiveness, efficiency, and satisfaction is also a challenge since getting access to end users can be difficult. Testing and improving the system design based on user feedback is a fundamental aspect of a User-Centered Design (UCD) approach. Without the end user, the iterative testing process cannot fully perceive the functionality and adaptability of the system in real-life environments [3].
- System adaptability to users with diverse motor and cognitive capabilities is a demanding task but fundamental when developing assistive device solutions that aim to be inclusive [4].
- Data privacy and ethical concerns are important non-technical challenges in the development of assistive systems. Dealing with personal and sensitive data, such as brain activity, brings safety and consent implications. Thus, is crucial to ensure compliance with data privacy policies and regulations by guaranteeing data encryption, secure storage protocols, and transparency [5].

To successfully implement an HCD, and comply with the guidelines, this project follows a sequence of five main stages: (A) Analysis of impaired people's daily routines to identify their needs and struggles; (B) Create personas and scenarios to comprehend which tasks are truly relevant and impactful in these subjects lives; (C) Requirements definition for the interface and interaction designs that enables users to interact with the environment; (D) Low-fidelity prototyping to test the system usability and to identify problems in the design; (E) Interface implementation and Wizard of Oz testing in healthy subject.

B. Design-by-proxy

Since the users may present difficulties in articulating and expressing their needs a design-by-proxy approach was implemented to address this issue. This method ensures that

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crucial information is gathered from health professionals who have direct contact with this population and understand their feelings, daily challenges, and necessities.

The first stage involved the use of visual maps to guide health professionals and caregivers in comprehending the project's main goal. Then a questionnaire was developed with specific questions that focused on three categories: (1) Identifying the daily struggles and primary challenges; (2) Determining the actions that the users most desire to perform independently; and (3) Understanding how professionals communicate with these users and the tools or technology accessible to them.

C. Personas & Scenarios

Personas are defined as fictitious representations of the end-users based on the profile data gathered during questionnaires. Additionally, scenarios are sets of events that project real situations from the personas' lives that are relevant to the design solution. Ultimately, the combined use of personas and scenarios guarantees that the interface is developed for real-life necessities that are fundamental to users' needs [8], [9]. Describing how the system interacts with personas to achieve their personal goals helps identify essential system functionalities and ensures that developers have a human perspective and approach to the design [8].

D. System Requirements

These personas and context scenarios support the definition of system requirements. Regarding this project, the requirements for the interface design were defined based on: (1) end-users' daily necessities; and (2) interface control system. The main motivation for impaired subjects, primarily those with locked-in syndrome, is to communicate, thus this interface had to provide proper communication. However, considering that these subjects can only communicate through their visual system, the design of the interaction and control of the interface had to be developed based on eye movements and brain activity.

Eye and brain control of the interface allows a natural, intuitive and hands-free interaction. However, the interface design must consider eye fatigue and discomfort [10] and prevent accidental selections due to involuntary eye movements. To ensure an adequate user interface design the 10 heuristics described by Nielsen [11] and by Zhang [12] on a medical perspective, allowed to conduct a structure evaluation focused on identifying crucial problems regarding the user control, system consistency, error prevention, and minimalist design.

E. Low-Fidelity Prototyping

A paper low-fidelity interactive model was created so that health professionals and caregivers could test the system interface and access whether it met their expectations and accessibility needs. This approach allowed a rapid and inexpensive iteration. This step was essential before implementing the interactive digital version of the interface.

F. System Implementation

This stage describes the interactive system implementation, focusing on the EEG signal decoding and the GUI.

1) SSVEP-BCI system: The paradigm control for the BCI system was the SSVEP [10][13]. EEG signals interpret the user intention based on specific brain frequencies, thus the actions are triggered according to the detected SSVEP signals from the user. Buttons blink at predefined frequencies (e.g., 6 Hz for "Yes" option, 12 Hz for "No" selection) to correspond with EEG signals captured during the BCI process. The subject focuses on the desired action button and the the blinking frequency of the button is detected by power analysis of the EEG signal.

EEG signal acquisition was performed using a one-channel wireless BITalino device [14] equipped with a small amplifier. BITalino connects via Bluetooth and acquires the EEG signal at a sampling rate of 100 Hz. The electrodes were placed on the scalp according to the 10-20 international system, Fp1 was the active electrode (positive), Fpz was the reference (negative) and the TP10 was the ground electrode. Although the active electrode is typically placed near the occipital region for SSVEP detection [15], technical challenges and hardware limitations resulted in excessive noise when testing the electrode in this area. As a result, for this study, the active electrode is located in the frontal region.

Regarding the EEG signal processing, a bandpass filter was applied to frequencies between 0.5 Hz and 30 Hz, to reduce noise and extract the band of interest. Then the Welch method was applied to calculate the signal Power Spectral Density (PSD) and identify significant peaks at the target frequencies (6, 8 and 12 Hz) that exceed a baseline threshold. The baseline corresponds to the mean power values for each frequency and is considered a significant peak when the power is at least twice the baseline. The detected power peaks are displayed in real-time in order for the developer to analyse the bar graph and understand if the system is operating correctly and accurately decoding the user intentions.

2) Graphical User Interface: The interface for the system was developed using Python's Tkinter library [16] and auditory feedback was achieved using pyttsx3 text-to-speech conversion library [17]. For the stimulus presentation for SSVEP detection, actions are selected through simple, large buttons that alternate at predefined frequencies between green and orange to provide good visual contrast. Layout specifications and design choices based on the identified requirements for the interaction design are described in more detailed in Subsection III-C.

III. RESULTS & DISCUSSION

The different results are discussed and presented in this section.

A. Questionnaires

Questionnaires were completed by six health professionals (one cardiopneumologist, two speech therapists, two nurses,

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and one doctor), as well as two caregivers. These professionals were able to address specific challenges and needs of the individuals based on direct contact with them and their environment. The word map in Fig. 4 presents the most common ideas and responses gathered during the questionnaires. Key terms such as 'Communicate', 'Gestures' and 'Support' reflect the great impact that allowing them to communicate could have in their lives.



Fig. 4. Word cloud from the questionnaires

The main findings showcase the lack of tools and technology accessible for professionals to communicate with individuals with severe impairments. Caregivers expressed their concern for these subjects since they reported feelings of loneliness and isolation due to their limited ability to communicate, which can contribute to emotional distress and cognitive decline.

B. Personas & Scenarios

1) Personas: The primary persona is a man who lives with Locked-in syndrome, and two additional Personas were included representing his most common interactions. These profiles were constructed based on the experience of health professionals and caregivers who dealt with subjects in similar conditions, and from the literature.

Primary Persona, Fig.5 (left), is 32 year-old Carlos Martins. Carlos was an active individual before being diagnosed with Locked-In Syndrome following a stroke. He has limited motor capabilities, restricted to eye movements. His main interests include outdoors activities in nature and appreciating beautiful landscapes. He often feels frustrated due to his inability to communicate with his family. His motivation relies on regaining some independence that could allow him to find a remote job and be financially independent.

The secondary Persona is a family member, 56 year-old Amália Correia, Fig.5 (midle), Carlos mother, she works a full-time job as a cashier at a local supermarket. Amália is a preoccupied mother who worries about Carlo's well-being and she struggles to balance work and caregiving responsibilities, leading to stress and exhaustion. She needs a user-friendly interface to help her assist Carlos remotely and that notifies her when Carlos is unwell.

Miranda Marques, Fig.5 (right), is a professional caregiver, she is 37 years old and she is responsible for assisting Carlos in his daily activities, personal care, and meal preparation, she is often the primary contact for his immediate needs. She is experienced working with impaired individuals and is passionate about discovering new assistive technologies that could promote Carlos's autonomy. Miranda visits on a fixed schedule, meaning Carlos may need assistance outside of her working hours.



Fig. 5. Personas: Carlos Martins, Primary Persona (left); Amália Correia, mom (midle); and Miranda Marques, careguiver (right) (Public domain images from freepike.com, retrived from [18]).

2) Scenarios: Scenario 1 describes Carlos sending a morning message to the family group chat. It's 8:00 AM, Carlos has just finished breakfast and he just wants to wish his family a good day. He opens his app by looking at the iPad, and then he selects the option "Messages". The voice assistant confirms his intentions by asking him if he wants to send to the family group a text. After his confirmation, he receives sound and visual feedback that the message has been sent. Moments later, Carlos sees the replies from his parents and feels happy and connected to his family.

Scenario 2 involves turning on the TV for entertainment. It's 4:00 PM and Carlos is bored and wants to watch some television. He opens his app and selects the option "Turn on the TV". The voice assistant asks him which of his top two favorite shows he would like to assist. After selecting the program, the TV turns on and Carlos feels happy to have autonomously selected a program to watch.

C. System Requirements

System requirements were defined based on the information gathered from the interviews and the scenarios so that the interface is context-aware and includes key design features that provide the ability for the users to communicate and be autonomous in daily simple tasks such as using the heuristics as guidelines and the foundation for the interface design.

The main goal of the actions implemented in the interface was to address the issues referenced in the questionnaires. Thus, users could chose between sending a message to their family group chat, adjust their bed for greater comfort, and select between their two favourite a TV programs for entertainment. The following items address and describe the implemented key features for the interface design.

1) Interface Design and Interaction: The design must be simple with a minimalist layout with a reliable decision-making process with both visual and auditory feedback. The system was developed specifically for Portuguese users, meaning all features were implemented in Portuguese.

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Simplicity and intuitiveness were prioritized to minimize cognitive demand. The interface layout is uncluttered and clean with big, clearly labelled buttons representing each action that the individual can perform. This design contributes to an easy user flow, consistently returning to the home screen after each task or attempt. To prevent accidental selections, the subject has a two-stage confirmation process, where after selecting an action, for it to take place, the user needs to confirm the selected action, ensuring user intent and minimizing error occurrence (reinforcement decision-making). Loading buttons were also implemented so that the subject had visual feedback, allowing users to undo selections. Additionally, the interface provides mixed sensory feedback, through visual cues and voice interaction that contributes to user engagement by asking for action confirmation and delivering immediate feedback after a task is performed or cancelled, enhancing clarity on the system status.

2) Safety: Reliability is a major concern when developing assistive systems, to address this issue, the system includes an emergency button represented by a big, red button blinking at 8 Hz on the home screen of the interface. Additionally, users are limited to simple tasks that enhance comfort and entertainment, reducing the risk of triggering complex or potentially dangerous actions.

3) Control Mechanism: The main goal of the interface was to enable hands-free control, eliminating any physical interaction, and make it accessible to all users. Given that end users may have different cognitive capabilities, minimizing cognitive load was essential. Furthermore, hardware limitations regarding the EEG acquisition system could affect the signal quality and consequently system usability. Hardware constraints must be considered and tested before evaluating system usability with end users to prevent frustration and unmet expectations. Therefore, to assess the acquisition limitations the EEG setup was tested by the developers, where a simple frame with 4 buttons blinking at different frequencies were tested to understand if the system was able to detect which button the subject was staring at, Fig. 6.

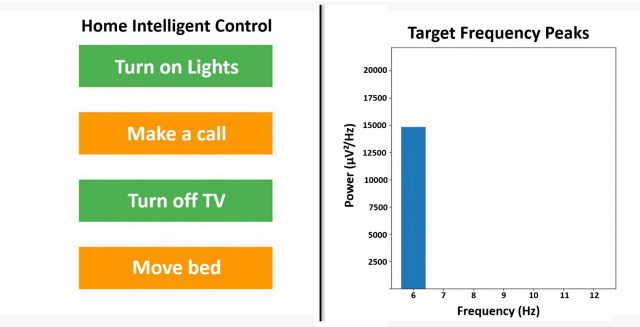


Fig. 6. SSVEP-BCI testing for interface control.

This test achieved poor performance, indicating low reliability for system control. The subjects struggled to focus on one single button when all were blinking simultaneously, and eye fatigue was reported after 5 minutes. Thus, to improve accuracy, the number of buttons per frame has been reduced to two, since distinguishing between four options is challenging.

D. Prototype Evaluation

After defining the requirements the interface design was implemented and the hand-drawn sketches allowed design adjustments and the integration of new features based on the feedback provided by the four participants who tested the prototype, two caregivers who evaluated the interface from an end-user perspective and two engineers who focused more on the technical aspects (Fig. 7). Additionally, the think-aloud approach provided crucial information about the interaction issues and guided the necessary refinements to enhance the system before the practical implementation.



Fig. 7. Low-fidelity prototype evaluation.

As a result, this low-fidelity approach identified some initial issues. At certain stages of selection, the subjects were unable to revert the option. To mitigate this problem, the interface was re-designed to return to the main screen after 30 seconds of inactivity. This allowed a reduced number of buttons per frame while ensuring a key functionality for system user flow.

E. Practical Implementation

After testing the paper prototype, the graphical implementation of the design was implemented in Python and the Wizard of Oz methodology was applied to simulate a perfectly functional interface control, which allowed to test the interface user flow and monitor the SSVEP responses to the visual stimulus. The main goal was to: (1) access system functionality and (2) test EEG signal decoding performance.

The systems was tested on 2 subjects that described the interface and the selection process while the developers selected the options for them making him believe he was the one selecting the options. Subjects could navigate through the application and select desired actions, such as turning on the TV, sending messages, adjusting bed positions, and activating emergency protocol (Fig. 8).

The emergency button activation proved inefficient, as users needed to focus on selecting a button which can be challenging under stress. As a mitigation strategy, integrating an eye-tracking device, such as Tobii Eye Tracker 5 [19], that detects eye movement behavior is crucial to obtain a pleasantly usable interface. With this strategy, the system activates the emergency option when users blink uncontrollably at any frame of the interface without having to focus on a specific task. Additionally, test subjects reported discomfort

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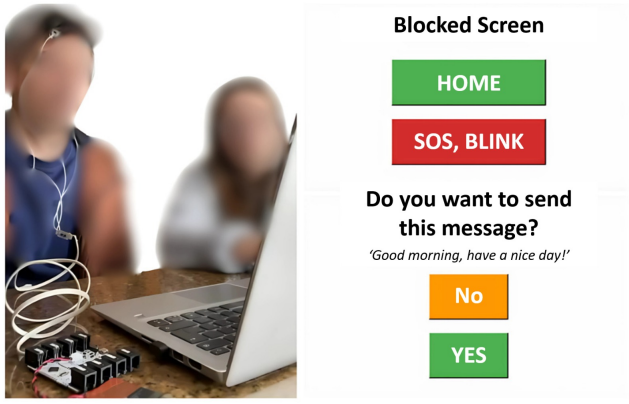


Fig. 8. Testing the interface user flow with Wizard of Oz, and examples of the interface minimalist design.

from being constantly stimulated with blinking buttons. To address this issue the eye tracker allows the interface to have a static locked home screen, which unlocks with eye gaze. Afterwards, the main screen, with blinking buttons, is actively controlled using the SSVEP paradigm and the eye movement.

In terms of prototype design this addition does not change crucial elements but adapts the control method. By allowing some frames to be static and selected with the eye interaction, reduces fatigue during interface navigation and enhances control accuracy by combining EEG and eye signals (dual-input control). Additionally, communication can be optimized by including a graphical keyboard, allowing users to spell their thoughts and send personalized messages, since this primary prototype allows the users to send simple predefined messages.

To achieve a functional system, real-time SSVEP-EEG decoding performance needs to be improved. Therefore, classification algorithms such as the training-free, Online Adaptive Canonical Correlation Analysis (OACCA) algorithm, proposed by Wong et al. [13], or the High-Order Temporal Convolutional Neural Network (HOT-CNN) model proposed by Yang et al. [10] should be tested to implement a system with real-time control.

Finally, one of the subjects reported that in situations with poor lighting conditions, the system could not be operated, this issue should be addressed in future design iterations.

IV. CONCLUSION

This study highlights how a human-centered design approach influences the envision and development of assistive systems. Engaging healthcare professionals in the design process ensures that the system interaction aligns with clinical needs with real-world applicability. This framework provides guidelines for developing BCI-driven assistive technologies. Future work will focus on incorporating the eye tracker device for reliable interface control, validating this approach with larger user groups including end-users, exploring the potential of integrating keyboards for expressive communication, implement different EEG decoding algorithms, and understanding if acquiring other bio-signals could improve control accuracy.

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